

Multistage Feedback-Driven Causal Discovery from Textual Data with Large Language Models

Juntao Yang
HFUT¹
Hefei, China
juntaoyang@mail.hfut.edu.cn

Dayuan Cao
HFUT¹
Hefei, China
dayuancao@hfut.edu.cn

Kui Yu*
HFUT¹
Hefei, China
yukui@hfut.edu.cn

Xiang Wang*
NUDT²
Changsha, China
xiangwangcn@nudt.edu.cn

Jing Yang
HFUT¹
Hefei, China
jing.yang@hfut.edu.cn

Lin Liu
AU³
Adelaide, Australia
lin.liu@adelaide.edu.au

Jiuyong Li
AU³
Adelaide, Australia
jiuyong.li@adelaide.edu.au

Abstract

Learning causal relationships from unstructured textual data can significantly enhance decision-making capabilities and the robustness of AI inference. For example, extracting the underlying causal factors that influence consumer experience or product reputation from online reviews could provide valuable information to improve product recommendation strategies. However, existing mainstream data-driven causal discovery (CD) methods are primarily designed for structured numerical data, posing substantial challenges for application to textual data. Although large language models (LLMs) exhibit strong semantic understanding abilities, they still struggle to uncover complex causal relations among variables from such textual data. To address this gap, we propose a novel multistage feedback-driven CAusal diScovery from Textual data with LLMs (CAST). The core innovation lies in a self-reinforcing CD pipeline: First, it learns a set of low-redundancy, high-consistency causal factors through multi-round random sampling and semantic clustering. Then based on the learned causal factors, it transforms unstructured text into high-quality structured data via multi-round value assignment and entropy-based filtering. Finally, it presents a novel Local Label Chaos Factor (LLCF) to identify model-anomalous samples for the iterative discovery and validation of latent causal factors. Experimental results demonstrate that our framework can effectively identify causal relationships from unstructured textual data and significantly outperforms existing baseline methods in terms of both accuracy and robustness.

CCS Concepts

• Information systems → Data mining.

*Corresponding author (Kui Yu, Xiang Wang).

¹The Key Laboratory of Knowledge Engineering with Big Data (the Ministry of Education of China), Hefei University of Technology, Hefei, Anhui 230601, China. ²National University of Defense Technology, Changsha, China. ³Adelaide University, Adelaide, Australia.



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792633>

Keywords

Causal discovery; Unstructured textual data; Large language models; Feedback optimization; Causal factor extraction

ACM Reference Format:

Juntao Yang, Dayuan Cao, Kui Yu, Xiang Wang, Jing Yang, Lin Liu, and Jiuyong Li. 2026. Multistage Feedback-Driven Causal Discovery from Textual Data with Large Language Models. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792633>

1 Introduction

Causal discovery (CD) [5, 7, 18] aims to identify the underlying causal relationships between variables from observational data and has emerged as a vital research direction in machine learning and data mining. In real-world scenarios, a vast amount of observational data containing latent causal information exists in the form of unstructured, user-generated texts [13]. The ability to directly and effectively extract causal relationships from such unstructured texts would offer valuable insights to enhance business decision-making and strengthen the robustness of AI inference [16, 21]. For example, in online healthcare communities, the identification of key causal drivers associated with a particular disease from medical histories and lifestyle records not only facilitates more accurate prediction of disease risk, but also provides explainable causal evidence to support personalized health interventions. Similarly, mining causal factors that influence consumer satisfaction and product reputation from online customer reviews can allow e-commerce platforms to better align with consumer preferences through optimized product presentation and recommendation strategies.

However, traditional data-driven CD methods (e.g. CMB [6], PC [17], and PCSL [7]) are designed primarily for structured numerical data, which poses significant challenges when applied to textual data [3, 13]. Recently, large language models (LLMs) have emerged as a promising alternative to identify causal relations from text [12, 25]. Although LLMs have shown encouraging results in CD, they mainly focus on extracting pairwise causal relations between from text and often struggle to uncover more complex causal graphs between multiple variables [1, 8]. Consequently, LLMs are still commonly used in conjunction with traditional CD methods to extract causal relationships from structured numerical data [5, 25]. For example, they employ LLMs to initialize causal structures as priors or refine causal graphs generated by traditional CD approaches.

Then mining causal relations directly from unstructured textual data remains an open research problem.

To address this challenge, Liu et al. [13] proposed the COAT algorithm, which represents a pioneering effort to leverage LLMs for directly mining causal relations from unstructured textual data. The COAT algorithm primarily involves three steps: first, it extracts potential causal factors from unstructured text; second, it constructs a structured numerical data set based on these factors; and finally, it applies traditional CD algorithms to identify causal relations among the factors within the structured numerical data set. However, this approach is limited by an imprecise and unstable mechanism for extracting causal factors, as well as a non-systematic pipeline for converting unstructured text into structured numerical data. To enable end-to-end learning of causal relations directly from raw unstructured text and to overcome the limitations of existing methods, several key challenges remain:

- How to automatically and accurately extract semantically consistent and stable causal factors from unstructured text;
- How to accurately assign values to the identified factors based on textual content to generate a high-quality, low-noise structured numerical dataset;
- How to iteratively identify potential latent causal factors to enhance the completeness of the final causal factor set;
- How to construct benchmark data and evaluation metrics to assess CD methods from textual data.

To address the challenges, in this paper, we propose a multi-stage feedback-driven framework for CD from unstructured textual data using LLMs. Our main contributions are summarized as follows:

- We propose a robust factor extraction method by integrating LLMs with multi-round random sampling and semantic clustering for identifying potential causal factors in text.
- We design a factor assignment approach that employs multi-round LLM assignment with voting mechanisms for the generation of high-quality structured numerical data.
- We propose a novel local label chaos factor to design a feedback optimization strategy to detect hard-to-explain samples that may contain latent causal factors.
- We develop a method capable of generating synthetic textual data and corresponding evaluation metrics for the assessment of CD methods in unstructured textual data.

2 Related Work

Existing research on LLM-based causal discovery can be broadly divided into two main types of approaches [14, 22, 25]: the first type aims to learn complete causal graphs from structured numerical data, while the second focuses on identifying pairwise causal relationships between user-defined variables directly from text.

Methods in the first category typically learn reliable causal graphs from structured numerical data by confining the causal discovery capability of LLMs to an auxiliary role through integrating LLMs with traditional causal discovery algorithms (e.g. PC and FCI algorithms). A pioneering work in this line is *chatPC* proposed by Cohrs et al. [4], which reformulates conditional independence tests as natural language prompts and employs LLMs as statistical oracles to guide the PC algorithm. Zhang et al. [28] further developed an interpretable causal auditing framework where LLMs

validate and explain causal discovery results to enhance credibility. Long et al. [15] proposed a refined strategy that uses pairwise prompts and voting mechanisms to derive reliable causal priors which are integrated into scoring functions such as the BDeu score. Although these methods enhance the effectiveness of traditional causal discovery methods by leveraging LLMs, they rely on structured numerical data [2].

In contrast, the second category of methods treats LLMs as knowledge bases, focusing on extracting pairwise causal relationships between user-defined variables directly from text without using structured numerical data [20]. Typically, given textual input or variable descriptions, researchers employ prompt engineering to elicit causal pairs (e.g., “smoking → lung cancer”) from LLMs. For instance, Willig et al. [24] were the first to explore LLMs to identify pairwise causal relations from text, while Kıcıman et al. [10] demonstrated that directly querying LLMs can infer causal directions in a zero-shot setting. To improve performance, subsequent studies have introduced strategies such as contextual examples [11], chain-of-thought reasoning [23], and causal narratives embedded in natural language prompts [29]. Despite these advances, studies have indicated that while such methods perform reasonably well in identifying pairwise causal relationships, their reliability remains highly questionable [9]. Common issues include hallucinations, instability, and lack of verifiability. As noted by Zecevic et al. [27], LLMs may simply “rephrase” causal knowledge memorized during training, rather than performing genuine causal reasoning. Tu et al. [19] showed that LLMs are not able to infer causal graphs involving multiple variables from textual data. To address the limitations, Liu et al. [13] proposed the COAT algorithm, which uses LLMs to mine causal relations between multiple variables directly from text. Despite this progress, COAT still faces several critical challenges. In this paper, we propose to tackle these challenges.

3 Proposed Method

In the section, we propose a multistage feedback-driven Causal discovery from Textual data with LLMs, named CAST. As shown in Figure 1, CAST comprises three components: 1) *robust causal factor extraction* (Section 3.1), 2) *factor assignment and numerical data construction* (Section 3.2), and 3) *causal discovery and feedback optimization* (Section 3.3). The three components are integrated into a single end-to-end framework.

3.1 Robust Causal Factor Extraction Module

This module aims to extract a set of high-quality candidate causal factors from unstructured text to accurately represent the original semantic content. For example, consider the following authentic watermelon review: “*The watermelon was much smaller than expected, with pale flesh color indicating incomplete ripening, and almost no sweetness. Very disappointed - would not recommend.*” This concise yet informative statement implicitly contains multiple evaluation dimensions including size, ripeness, and taste, all of which may serve as key causal factors influencing product sales. In general, to accurately represent the semantic information of the original text, an ideal factor set satisfies three key properties: 1) low redundancy (by merging semantically similar factors), 2) low noise (by filtering out irrelevant factors), and 3) strong consensus (by retaining core

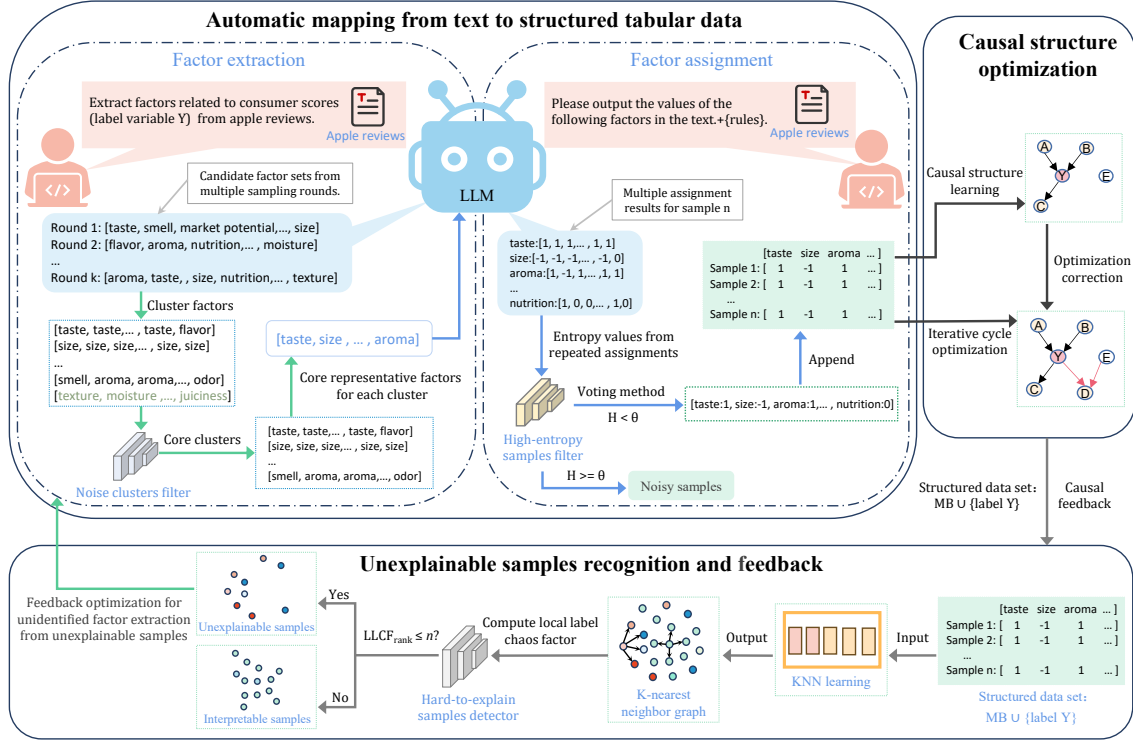


Figure 1: The overall framework of CAST.

concepts that are universally present across most samples and can be consistently identified by LLMs). To achieve this goal, we propose a robust extraction method consisting of iterative sampling, semantic clustering, and noise filtering stages.

1) Iterative sampling to extract potential causal factors. We employ LLMs for multiple rounds of stochastic sampling to mine potential causal factors from text. We first partition the text dataset $\mathcal{D}$ into groups based on labels, then construct K diverse data sample subsets $\{S_1, S_2, \dots, S_K\}$ through independent random sampling. For each subset S_i , we query LLMs using a specialized prompt P_{factor} to obtain causal factor set F_i . The factors extracted from each subset are merged: $\{F_1, F_2, \dots, F_K\} = \bigcup_{i=1}^K \text{LLM}(P_{\text{factor}}, S_i)$. This multi-round approach ensures diversity in the factor extraction process while providing the foundation for assessing consensus factors (factors that are universally present across most samples).

2) Semantic clustering to reduce factor redundancy. With the identified causal factor sets, we perform semantic embedding and clustering for merging redundant factors. Specifically, we first combine all identified causal factor sets into an initial set F_{total} and encode each factor $f \in F_{\text{total}}$ to a high-dimensional vector using a pre-trained sentence embedding model and get the embedding set E of F_{total} . We then perform agglomerative hierarchical clustering on E , setting the number of clusters to M , and obtain clusters $\{C_1, C_2, \dots, C_M\} = \text{AgglomerativeClustering}(E, M)$. The cluster count M is estimated based on the scale of potential factors initially extracted by the LLM, ensuring it aligns with a reasonable range of latent factors present in the text and is not set arbitrarily.

Each cluster C_j represents a potential causal concept containing semantically similar factors and can exhibit varying quality levels. Some contain inconsistent or irrelevant causal factors.

3) Noise filtering to eliminate irrelevant clusters. To evaluate cluster quality and identify noise clusters, we present the following two complementary metrics.

i) Normalized Intra-class Conditional Entropy ($H_{\text{norm}}(C_j)$). The metric quantifies the semantic chaos within a cluster by measuring the disorder in the factor label distribution and is defined as follows.

$$H_{\text{norm}}(C_j) = \begin{cases} 0 & \text{if } |U_j| = 1, \\ -\frac{\sum_{f \in U_j} p(f) \log_2 p(f)}{\log_2 |U_j|} & \text{if } |U_j| > 1, \end{cases} \quad (1)$$

where U_j is the set of distinct factors in cluster C_j , $|U_j|$ is the size of this set, and $p(f) = n_f/N$ is the relative frequency of factor f in cluster C_j (n_f is the occurrence count, N is the total count). When all factors in a cluster are the same one, $H_{\text{norm}} = 0$ holds. The value of H_{norm} approaching 1 indicates uniform label distribution (high chaos). High chaos clusters typically represent noise clusters containing inconsistent LLM outputs or irrelevant concepts.

ii) Intra-class Dominant Factor Cohesion (IDFC). This metric measures the similarity between a cluster's actual factor distribution and an ideal distribution centered around the dominant factor f_j^* (the highest-frequency factor in the cluster), evaluating how closely factors cluster around the core concept. The actual distribution P_{C_j} is derived from factor frequencies in a cluster:

$$P_{C_j}(f) = \frac{n_f}{N}, \quad \forall f \in C_j. \quad (2)$$

The ideal cohesive distribution Q_{C_j} is a hypothetical distribution where the dominant factor f_j^* occupies absolute dominance (with probability mass a , where $a = 1$ ideally, indicating all instances are the same factor), while remaining factors share the residual probability mass equally:

$$Q_{C_j}(f) = \begin{cases} a & \text{if } f = f_j^*, \\ \frac{1-a}{|U_j|-1} & \text{otherwise.} \end{cases} \quad (3)$$

IDFC is computed by measuring the divergence between P_{C_j} and Q_{C_j} relative to their average $M = \frac{1}{2}(P_{C_j} + Q_{C_j})$ using KL divergence and mapping it to $[0, 1]$:

$$\text{IDFC}(C_j) = 1 - \frac{1}{2} \left[D_{\text{KL}}(P_{C_j} \parallel M) + D_{\text{KL}}(Q_{C_j} \parallel M) \right], \quad (4)$$

where $D_{\text{KL}}(A \parallel B) = \sum_i A(i) \log \frac{A(i)}{B(i)}$ is the Kullback-Leibler divergence. When the actual distribution P_{C_j} exactly equals the ideal distribution Q_{C_j} , IDFC = 1, indicating maximum cohesion and highest cluster quality. A lower value of IDFC indicates more dispersed distributions, suggesting insufficient representation by the dominant factor and potential noise clusters. We establish thresholds α and β for the two metrics respectively. A cluster C_j is identified as a noise cluster and filtered out if it satisfies:

$$H_{\text{norm}}(C_j) > \alpha \quad \text{and} \quad \text{IDFC}(C_j) < \beta. \quad (5)$$

Finally, in the core factor set generation phase, for each high-quality cluster C_j that passes filtering, we select its dominant factor f_j^* (the highest-frequency factor representing consensus across multiple sampling rounds) as the unique representative of that semantic concept. The final core causal factor set V thus possesses low redundancy, low noise, and high consensus, establishing a reliable foundation for subsequent constructing numerical data.

3.2 Robust Factor Assignment and Dataset Construction Module

Upon obtaining a high-quality set V of core causal factors, the primary objective of this module is to transform each sample i in textual dataset D into a structured numerical data sample $\mathbf{i} = (i_1, i_2, \dots, i_m)$, where $m = |V|$ and i_j represents the numerical value of a causal factor j for the text sample i . This transformation process must simultaneously fulfill three critical objectives: high fidelity (each numerical value accurately reflects the original semantic content), high efficiency (controlling LLM query costs), and low noise (identifying and handling uncertain assignments). To achieve these goals, we design a robust factor assignment framework consisting of three components: multi-round LLM assignment with caching mechanism, noise identification and filtering via information entropy, and structured numerical dataset integration. Specifically, the first component is for mitigating potential random errors when using single-round queries, the second component aims to filter out text with ambiguous descriptions or insufficient information, and the last one is to construct a high-quality numerical dataset.

1) Multi-round LLM assignment with caching mechanism. For a text sample i and a causal factor $j \in V$, a single LLM query may produce biased or erroneous results due to prompt randomness.

To enhance the stability of assigned values of j , we execute up to N_{iter} independent assignment queries for each (i, j) pair:

$$R_{ij} = \{j_i^1, j_i^2, \dots, j_i^{N_{\text{iter}}}\} = \bigcup_{k=1}^{N_{\text{iter}}} \text{LLM}(P_{\text{assign}}, i, j), \quad (6)$$

where P_{assign} is the assignment prompt, k is the round index, and j_i^k represents the value for factor j in sample i at round k . All assignment results are cached in a result pool R_{ij} . To improve efficiency and reduce unnecessary LLM calls, we design an intelligent early termination strategy with dual criteria. After each assignment round, the system dynamically examines the cached results.

i) Consensus-based termination criterion. This criterion leverages the mathematical guarantee that any value reaching an absolute majority (more than $N_{\text{iter}}/2$ occurrences), it cannot be overtaken by any other value in subsequent rounds. This insight allows us to safely terminate queries once a value satisfies $\sum_{k=1}^t \mathbb{I}(j_i^k = v) > N_{\text{iter}}/2$.

ii) Stability-based termination criterion. This criterion addresses scenarios in which LLM outputs stabilize into a consistent pattern—even absent an absolute majority—by triggering termination when consecutive τ rounds produce identical results, i.e., $j_i^{t-\tau+1} = j_i^{t-\tau+2} = \dots = j_i^t$, where τ represents the stability window (typically $\tau = 2$), this indicates that the LLM's interpretation has reached a steady state. Such consistency suggests that the textual description of the factor is sufficiently clear to produce stable assignments, and continuing queries would likely yield diminishing returns. Conversely, if the results continue to be unstable, this suggests that the text description may be ambiguous, requiring more rounds of queries to capture the uncertainty and subsequent quality assessment steps. The actual termination round T_{ij} is formally defined as:

$$T_{ij} = \min \{t \leq N_{\text{iter}} : \text{Consensus}(R_{ij}, t) \vee \text{Stability}(R_{ij}, t, \tau)\}. \quad (7)$$

If neither condition is satisfied after completing all N_{iter} rounds, the system proceeds to the entropy-based evaluation phase. This comprehensive approach ensures that we efficiently handle both clear-cut cases through early termination while reserving more sophisticated analysis for ambiguous scenarios.

2) Noise identification and filtering via information entropy. For (i, j) pairs that complete all N_{iter} rounds without triggering early termination criteria, the assignment results exhibit high uncertainty, potentially indicating textual ambiguity or noisy samples. To achieve our low-noise objective, we employ information entropy to quantify this uncertainty and perform systematic noise filtering. The information entropy is computed over the N_{iter} values in the cache pool:

$$H_{ij} = - \sum_{v \in U_{ij}} p(v) \log_2 p(v), \quad (8)$$

where U_{ij} denotes the set of distinct values observed across all N_{iter} rounds, and $p(v) = \frac{n_v}{N_{\text{iter}}}$ represents the empirical frequency of value v , with n_v being its occurrence count.

The entropy value H_{ij} directly measures the uncertainty in LLM assignments: higher values indicate greater inconsistency in the model's outputs, suggesting unreliable assignments that may

stem from ambiguous textual descriptions. We establish an entropy threshold θ to automatically identify and handle such cases:

$$i_j = \begin{cases} 0 & \text{if } H_{ij} > \theta, \\ \text{Majority}(R_{ij}) & \text{otherwise,} \end{cases} \quad (9)$$

when $H_{ij} > \theta$, we mark i_j as missing (encoded as 0), flagging sample i as noisy for factor j . Otherwise, we apply majority voting to determine the final assignment. This entropy-based filtering ensures the construction of a structured dataset with high internal consistency and reliability.

3) Structured numerical dataset integration. We iterate through all samples $i \in D$ and factors $j \in V$, applying the aforementioned procedures systematically. The final structured dataset X integrates all reliable assignments and properly handled noisy samples. Specifically, X is an $n \times m$ matrix where the element in the k -th row and j -th column, denoted by $i_j^{(k)}$, represents the value of factor j for the k -th sample. Here, $n = |D|$ is the total number of text samples, and $m = |V|$ is the size of the core factor set. This structured numerical dataset X provides a solid foundation for subsequent causal graph learning using existing CD algorithms.

3.3 Causal Discovery and Feedback Optimization Module

By the obtained dataset X defined on the factor set V , we employ a well-known causal discovery algorithm named FCI (any other state-of-the-art causal discovery algorithms can be used here) to learn a causal graph from X , i.e., $G_0 = \text{FCI}(X)$. Due to the complexity of unstructured textual data, some key causal factors are not identified by the preceding modules and are not added to V .

To mine the unidentified causal factors, COAT partitions data samples into multiple clusters using the K-means method, then selects the cluster with the maximum conditional entropy of ground-truth labels as the hard-to-explain sample set which may contain unidentified causal factors. But this method has two limitations: 1) it struggles to precisely identify hard-to-explain samples, as high-entropy clusters may contain a substantial number of explainable samples; 2) the preset number of clusters severely leads to the instability of identified hard-to-explain samples.

To address the limitations, we propose a feedback optimization module based on a novel Local Label Chaos Factor (LLCF), consisting of three steps: 1) hard-to-Explain Sample Identification via Local Label Chaos Factor, 2) progressively uncovering latent causal factors, and 3) end-to-end iterative optimization. This module is motivated by the following insight: a sample that is well represented with the current causal factor set V should demonstrate label consistency with its neighboring samples. Conversely, if the local neighborhood of a sample exhibits a chaotic label distribution, this suggests that the set V lacks sufficient explanatory power for that sample—often because certain underlying key causal factors have not yet been identified. For example, consider a low-satisfaction review: “The watermelon looked great - large and fresh - but it tasted absolutely terrible!” Under the current factor set {“size”, “freshness”}, its nearest neighbors (which are also large and fresh) are mostly high-satisfaction samples. This results in low local label consistency, so the sample is identified by LLCF as a hard-to-explain sample. By re-extracting factors from this specific sample, we recover the

missing key factor “taste.” Thus, compared with COAT, LLCF advances from “coarse cluster-level selection” to “precise sample-level localization”, enabling more accurate and efficient factor discovery.

1) Hard-to-explain sample identification via local label chaos factor. To identify the hard-to-explain samples within the context of the current factor set V , we employ the Markov Blanket (MB) from causal discovery.

Specifically, in a causal graph, the MB of a variable Y is formally defined as the unique set comprising its parent nodes (direct causes), child nodes (direct effects), and spouse nodes (other parents of its children), i.e., $\text{MB}_Y = \text{Pa}(Y) \cup \text{Ch}(Y) \cup \text{Pa}(\text{Ch}(Y) \setminus Y)$. Figure 2 shows an example of an MB in a causal graph and the MB of Y consists of A and B (parents), C and D (child), and E (spouse). Here, Y could represent a(n) outcome/label variable of interest, such as a specified disease in medical records or consumer satisfaction in product reviews. The MB set possesses a good theoretical property:

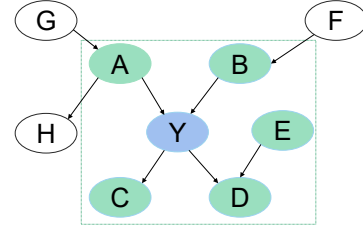


Figure 2: An example of an MB of Y in a causal graph.

conditioning on MB_Y , all other variables are conditionally independent of the variable Y . This makes MB_Y the minimal feature set containing all necessary information for predicting Y [26]. Based on this, we project the samples from X into the feature subspace defined by MB_Y to perform accurately K-nearest neighbor search. For each sample i , this obtains its k nearest neighbors $N_k(i)$ and their corresponding labels. To better quantify the anomaly degree of a sample, we introduce a distance-based weighting mechanism when evaluating the label chaos within $N_k(i)$. This design is grounded in the rationale that the influence between samples is distance-dependent, and thus label discrepancies from distant neighbors are more significant than those from closer ones.

Then based on the exponential decay law $A(d) = A_0 e^{-\lambda d}$, we define the Repulsion Potential (RP) of a neighboring sample j toward the central sample i according to their distance, quantifying the local anomaly degree caused by neighbor j to center i :

$$\text{RP}_{ij} = e^{-\gamma d_{ij}}, \quad (10)$$

where d_{ij} is the Euclidean distance between samples i and j , and γ is the decay coefficient.

To facilitate comparison of local label chaos across different samples, we normalize the repulsion potential of the k nearest neighbors for each sample, obtaining the weight coefficient w_{ij} :

$$w_{ij} = \frac{\text{RP}_{ij}}{\sum_{j' \in N_k(i)} \text{RP}_{ij'}}. \quad (11)$$

This weight satisfies the normalization condition $\sum_{j \in N_k(i)} w_{ij} = 1$. Subsequently, we define the local label chaos factor (LLCF) of

sample i as the sum of repulsion potentials from neighbors with differing labels within its k -neighborhood:

$$\text{LLCF}(i) = \sum_{j \in N_k(i)} w_{ij} \cdot \mathbb{I}(y_j \neq y_i), \quad (12)$$

where $\mathbb{I}(\cdot)$ is the indicator function that equals 1 when $y_j \neq y_i$ and 0 otherwise. Within our CD framework, a sample with high LLCF scores indicates that the current factor set fails to explain why the particular sample appears in its current positions. LLCF acts as a "detector" that precisely locates the blind spots of the current model, then guiding subsequent factor mining procedures to focus on regions most likely to contain unidentified factors. We rank all samples by their LLCF scores and select the top- n samples to form the hard-to-explain sample set D_{hard} . These samples represent the most significant "blind spots" in the current causal graph.

2) Progressively uncovering latent causal factors. Upon obtaining the hard-to-explain samples D_{hard} , we put them into the robust causal factor extraction module described in Section 3.1 to guide LLMs to focus on identify potential factors that "can explain why these specific samples behave so differently from their neighbors". This process yields a candidate set of new factors $V_{\text{candidate}}$. However, not all new identified factors are correct and necessary. To avoid introducing redundant or incorrect factors, we design a rigorous validation strategy. For each candidate factor $v_{\text{cand}} \in V_{\text{candidate}}$, we employ the factor assignment module in Section 3.2 to obtain its values across all samples, producing an extended dataset $X_{\text{extended}} = [X, v_{\text{cand}}]$.

Based on the extended dataset, we recalculate the LLCF values for all samples. If the new factor v_{cand} indeed captures unidentified causal factors, it should effectively clarify the originally chaotic local neighborhood relationships, then significantly reduces the average LLCF value across the entire dataset. We select the candidate factors that achieve the maximum reduction in average LLCF and integrate them into the core factor set via $V_{\text{new}} = V_{\text{current}} \cup \{v_{\text{cand}}^*\}$.

3) Closed-loop iterative Optimization and its termination conditions. Based on newly discovered factors, we re-perform causal discovery using the FCI algorithm based on the updated factor set V_{new} and the reconstructed structured dataset, and iteratively executes a closed-loop workflow, until any of the following termination conditions are satisfied: (1) performance convergence: The average LLCF falls below threshold ϵ ; (2) no effective new factors: In a given iteration, no candidate factor passes the utility test; (3) maximum iterations reached: The process reaches the predefined maximum number of iterations.

This feedback mechanism transforms the entire CD process into a data-driven, self-improving intelligent system, forming a closed-loop optimization mechanism of "factor extraction $\rightarrow$ factor assignment $\rightarrow$ causal graph learning $\rightarrow$ hard-to-explain sample identification $\rightarrow$ new factor extraction $\rightarrow$ causal graph updating", ultimately outputting a more explanatory and complete causal model G_{final} .

4 Experiment

This section systematically evaluates the performance of CAST with multiple dimensions, including factor extraction, factor assignment, and causal discovery. We conduct comprehensive testing using nine prominent language models in three datasets. Due to space

constraints, we present only representative experimental results in the main text, while the complete set of results is provided in Appendix B.

4.1 Experimental Setup

Datasets. To evaluate the effectiveness of CAST, we conduct experiments on three datasets spanning both synthetic and real-world scenarios. *AppleGastronome (AG)* [13] is a synthetic e-commerce platform containing consumer reviews about apple purchases; *LungDisease (LD)* is a synthetic medical text dataset, and its details are provided in Appendix A; *Watermelon (WM)* is a real-world review dataset collected from Walmart's e-commerce platform, consisting solely of human-written product reviews. Table 1 summarizes the statistics of these datasets.

Table 1: DESCRIPTION OF DATASETS

Datasets	Instances	Category	Domain
AppleGastronome	200	Synthetic	E-commerce
LungDisease	1300	Synthetic	Medicine
Watermelon	778	Real	E-commerce

Evaluation Metrics. To systematically evaluate the performance of the CAST, we designed a four-stage evaluation system comprising factor extraction, factor assignment, causal discovery, and causal effect estimation. This system places particular emphasis on assessing the learning performance of the Markov Blanket (MB) — since the MB represents the minimal set of causal features that provides optimal predictive power for the target variable, accurately identifying the MB not only helps reveal key causal factors influencing the target variable but also provides actionable causal insights for network platforms rich in textual data, thereby effectively enhancing decision quality. Based on this, we focus specifically on evaluating MB learning performance in both the factor extraction and causal discovery stages. The specific composition of the four evaluation stages is as follows: (i) factor extraction, assessing the number of correct Markov blanket (MB) factors that LLMs directly identify for the outcome variable; (ii) factor assignment, measuring the accuracy of the annotated values for these factors; (iii) causal learning, evaluating the number of MB factors identified by the algorithm and its ability to recover the true causal structure; and (iv) causal effect estimation, measuring the accuracy of treatment effect estimation using standard metrics including average treatment effect error, Pearson correlation, and PEHE (due to space constraints, the experimental results for this part are provided in Appendix B). **Baselines.** We compare CAST against two state-of-the-art methods: **DATA+CoT**, which feeds raw texts to the LLM with chain-of-thought prompting for MB identification; **COAT** [13], a recent framework that combines LLMs with causal discovery.

4.2 Experimental Results and Analysis

Our experiments focus on evaluating method performance across three key dimensions: (1) the ability to extract key causal factors within the target variable's MB directly from raw text (Table 2); (2) the structural characteristics of learned causal graphs (Figure 2); and

Table 2: Comparison of three pipelines (DATA+CoT, COAT, CAST) across datasets and LLMs for factor extraction, Factor Assignment, and Causal Feature Learning. Best results are in bold.

Dataset	LLM	Method	Factor Extraction			Factor Assignment		Causal Feature Learning		
			Recall	Precision	F1	Accuracy		Recall	Precision	F1
AG	Deepseek-r1:14b	DATA+CoT	88.00	55.41	68.00	/		40.00	44.44	42.11
		COAT	92.00	61.13	73.45	76.20		80.00	82.38	81.17
		CAST(ours)	94.00	65.02	76.87	82.33		84.00	85.50	84.74
	Deepseek-v3-0324	DATA+CoT	100.00	55.56	71.43	/		84.00	52.54	64.65
		COAT	100.0	55.40	71.30	91.34		80.00	81.25	80.62
		CAST(ours)	100.0	59.39	74.52	91.80		94.00	87.50	90.63
	Qwen3:32b	DATA+CoT	80.00	32.14	45.86	/		76.00	39.65	52.12
		COAT	84.00	49.31	62.14	89.46		68.00	79.00	73.09
		CAST(ours)	84.00	64.42	72.92	90.36		72.00	96.67	82.53
	Qwen-plus	DATA+CoT	100.00	52.50	68.85	/		76.00	45.60	57.00
		COAT	100.0	75.00	85.71	90.29		94.00	86.14	89.90
		CAST(ours)	100.0	80.95	89.47	90.43		100.0	100.0	100.0
	Gpt-oss:20b	DATA+CoT	93.33	40.46	56.45	/		86.67	37.89	52.73
		COAT	98.00	42.99	59.76	87.07		80.00	77.48	78.72
		CAST(ours)	100.0	51.64	68.11	89.90		82.00	80.38	81.18
LD	Deepseek-r1:14b	DATA+CoT	76.00	8.33	15.02	/		24.00	23.33	23.66
		COAT	74.00	47.13	57.58	90.70		74.00	67.23	70.45
		CAST(ours)	96.00	53.08	68.36	95.67		84.00	78.00	80.89
	Deepseek-v3-0324	DATA+CoT	88.00	48.89	62.86	/		76.00	47.82	58.70
		COAT	100.00	49.09	65.85	99.96		96.00	93.14	94.55
		CAST(ours)	100.00	50.00	66.67	99.96		98.00	88.33	92.91
	Qwen3:32b	DATA+CoT	100.00	44.09	61.20	/		88.00	67.84	76.62
		COAT	88.00	45.22	59.74	99.97		88.00	72.62	79.57
		CAST(ours)	98.00	49.44	65.72	99.83		98.00	96.67	97.33
	Qwen-plus	DATA+CoT	100.00	52.50	68.85	/		84.00	92.67	88.12
		COAT	98.00	52.70	68.54	100.00		98.00	98.00	98.00
		CAST(ours)	100.00	52.70	69.02	100.00		100.00	98.33	99.16
	Gpt-oss:20b	DATA+CoT	73.33	73.33	73.33	/		53.33	88.89	66.67
		COAT	92.00	41.47	57.17	99.02		92.00	77.74	84.27
		CAST(ours)	100.00	51.11	67.65	99.88		96.00	83.62	89.38
WM	Deepseek-r1:14b	DATA+CoT	43.33	35.56	39.06	/		20.00	22.00	20.95
		COAT	76.67	40.16	52.71	74.18		43.33	29.43	35.05
		CAST(ours)	83.33	51.61	63.74	77.34		73.33	76.79	75.02
	Deepseek-v3-0324	DATA+CoT	50.00	31.50	38.65	/		46.67	43.81	45.19
		COAT	76.67	30.27	43.40	73.39		13.33	8.00	10.00
		CAST(ours)	76.67	33.45	46.58	73.52		66.67	61.43	63.94
	Qwen3:32b	DATA+CoT	66.67	15.87	25.64	/		40.00	18.74	25.52
		COAT	66.67	18.72	29.23	81.44		66.67	44.03	53.03
		CAST(ours)	66.67	41.79	51.38	79.08		66.67	79.29	72.43
	Qwen-plus	DATA+CoT	53.33	25.67	34.66	/		50.00	36.67	42.31
		COAT	66.67	38.57	48.87	64.00		45.00	48.79	46.82
		CAST(ours)	70.00	49.68	58.11	67.05		53.33	64.10	58.22
	Gpt-oss:20b	DATA+CoT	80.00	25.50	38.67	/		66.67	26.33	37.75
		COAT	90.00	21.59	34.83	87.11		66.67	45.66	54.20
		CAST(ours)	96.67	60.31	74.28	88.13		90.00	82.24	85.95

(3) the accuracy of subsequent causal effect estimation (Appendix B).

4.2.1 Factor Extraction Performance. The factor extraction block of Table 2 reports the recall, precision, and F1-score for MB factor extraction across three datasets and different models. Experimental results show that CAST achieves the best recall in 14 out of 15 dataset-model combinations, with only one exception on the LD dataset using Qwen3:32b, where it is 2% lower than DATA+CoT. However, in this case, CAST still outperforms the second-best method by 4.22% in precision and 4.55% in F1-score. Notably, on the LD dataset with Deepseek-14b, CAST improves recall by 20%. Furthermore, CAST achieves the highest precision across all experiments on all three datasets. On the real-world e-commerce review dataset WM, CAST also demonstrates significant improvements: with Deepseek-14b, the F1-score increases from 52.71% under COAT to 63.74% (+11.03%); with Qwen3:32b, it rises from 29.23% to 51.38% (+22.15%). These results indicate that the proposed multi-round sampling and semantic clustering mechanism effectively addresses challenges

such as textual sparsity and weak causal signals in real-world scenarios, establishing a solid foundation for extracting meaningful causal factors from user reviews or medical texts. For example, in apple and watermelon reviews, CAST more comprehensively captures key sales influencers such as “taste,” “size,” and “freshness,” not only providing high-quality feature inputs for subsequent causal relationship learning but also offering more comprehensive data support for the design of practical marketing strategies.

4.2.2 Factor Value Assignment Quality. The factor assignment stage aims to transform identified causal factors into high-quality structured data (see the “Accuracy” column in Table 2). Experimental results show that CAST achieves the highest accuracy in 13 out of 15 dataset-model combinations. In the two cases where it does not achieve the best performance, the gap is minimal—for instance, on the LD dataset with Qwen3-32b, CAST is only 0.14% lower than COAT. Meanwhile, CAST brings significant improvements across multiple models. For example, on the AG and LD datasets with Deepseek-14b, accuracy increases by 6.13% and 4.97%, respectively.

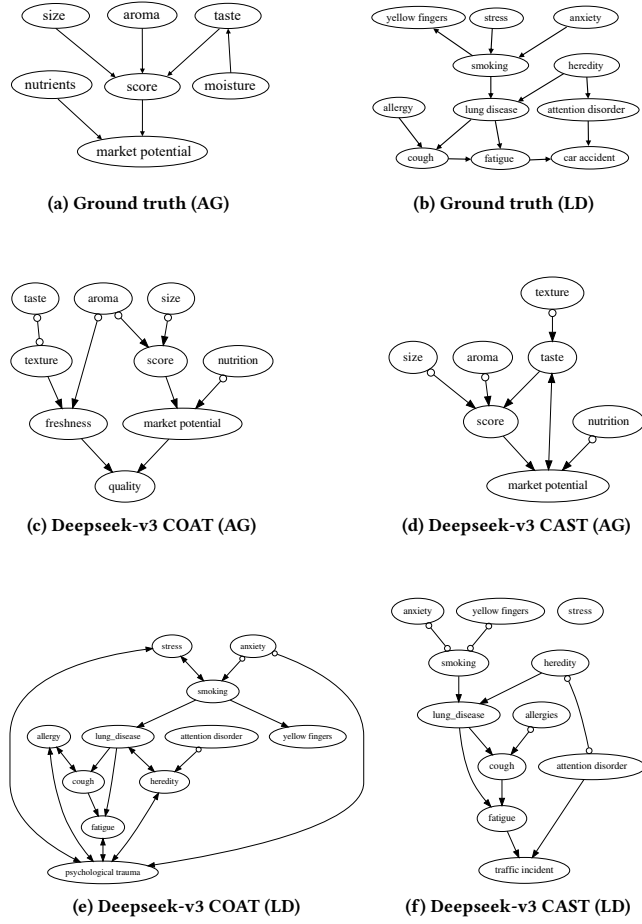


Figure 3: Causal graphs discovered by COAT and CAST(ours) across different LLMs.

Notably, the improvement is particularly pronounced in models with smaller parameter sizes. Taking Deepseek-8b on the AG and LD datasets as an example (experimental results can be found in Table 3 of Appendix B), accuracy improves by 8.16% and 17.95%, respectively. This indicates that CAST can effectively enhance assignment quality while maintaining low deployment costs, demonstrating strong potential for practical applications. These improvements are primarily attributed to the proposed multi-round assignment mechanism and entropy-aware noise filtering strategy, which work synergistically to significantly reduce annotation error rates. This enables the method to consistently output high-quality structured data even in real-world text scenarios characterized by irregular expressions and vague descriptions, providing a reliable foundation for constructing credible causal analysis tables from authentic internet texts, as well as establishing a data-driven basis for product optimization and user satisfaction mining in the e-commerce domain.

4.2.3 Causal Discovery Performance. To evaluate causal structure learning performance, we compare the causal graphs learned by

CAST and baseline methods on the AG and LD datasets. As shown in Figure 3, on the AG dataset, CAST (Figure 3d) accurately identifies all key factors affecting apple quality evaluation—including size, aroma, taste—and establishes clear causal paths to the "score". In contrast, while the COAT method (Figure 3c) captures most factors, it completely misses the "taste" dimension, which could lead to overlooking the impact of scent on consumer purchase decisions in practical marketing scenarios. On the medical-domain LD dataset, CAST demonstrates stronger capability in filtering out spurious causal relationships. For instance, the causal graph obtained by Deepseek-v3 with COAT (Figure 3e) contains medically unsubstantiated edges such as "psychological trauma → heredity" and "allergy → psychological trauma", whereas CAST (Figure 3f) generates a more concise and reliable causal structure that precisely focuses on clinically recognized paths like "smoking → lung disease" and "heredity → lung disease". This accurate identification of key medical factors provides more credible support for auxiliary diagnostic decision-making.

Quantitative results (from the "Causal Feature Learning" module of Table 2) further validate the superiority of CAST: in the MB identification task, CAST achieves optimal F1 scores in 14 out of 15 dataset-model combinations, with only one exception on the LD dataset using Deepseek-v3-0324 where it trails COAT by 1.64%. Furthermore, our method achieves performance improvements exceeding 10% in 8 combinations, with this advantage being particularly notable on the real-world WM dataset.

In summary, through its multi-stage collaborative optimization architecture, CAST more accurately identifies key causal paths during the iterative process, thereby significantly enhancing the reliability of automatically mining causal relationships from textual data. This provides more powerful support for e-commerce marketing strategy formulation and medical auxiliary analysis.

5 Conclusion

We propose CAST, an innovative framework for causal discovery from unstructured text, which systematically integrates large language models with causal inference techniques to effectively address the core challenge of mining latent causal relationships from textual data. Experimental results demonstrate that CAST significantly outperforms existing baseline methods across all evaluation stages, including factor discovery, structured data construction accuracy, and causal structure learning. Notably, our framework maintains excellent performance even when employing smaller-scale language models (with 8b-14b parameters) and when handling challenging real-world datasets, providing a practical and reliable solution for real-world deployment scenarios with limited resources. This research establishes a systematic methodology for directly extracting causal relationships from unstructured text. Looking ahead, we will continue to advance research on text-based causal discovery in online incremental learning and federated learning scenarios, further expanding the application boundaries of this technology.

Acknowledgments

This work was supported in part by the National Science and Technology Major Project of China (No.2021ZD0111801) and in part by the National Natural Science Foundation of China (No.62376087).

References

- [1] Ahmed Abdulaal, Adamos Hadjivasilou, Nina Montana-Brown, Tiantian He, Ayodeji Ijshakin, Ivana Drobniak, Daniel C Castro, and Daniel C Alexander. 2024. Causal modelling agents: Causal graph discovery through synergising metadata-and data-driven reasoning. In *ICLR*, Vol. 2024.
- [2] Taiyu Ban, Lyuzhou Chen, Derui Lyu, Xiangyu Wang, and Huanhuan Chen. 2024. Causal Structure Learning Supervised by Large Language Model.
- [3] Haoang Chi, He Li, Wenjing Yang, Feng Liu, Long Lan, Xiaoguang Ren, Tongliang Liu, and Bo Han. 2025. Unveiling Causal Reasoning in Large Language Models: Reality or Mirage?. In *NeurIPS*. Curran Associates Inc., Red Hook, NY, USA, Article 3064, 31 pages. ISBN: 9798331314385.
- [4] Kai-Hendrik Cohrs, Emiliano Diaz, Vasileios Sitokostantinou, Gherardo Varando, and Gustavo Camps-Valls. 2023. Large Language Models for Constrained-Based Causal Discovery. In *AAAI 2024 Workshop on "Are Large Language Models Simply Causal Parrots?"*.
- [5] Huaming Du, Yujia Zheng, Baoyu Jing, Yu Zhao, Gang Kou, Guisong Liu, Tao Gu, Weimin Li, and Carl Yang. 2025. Causal Discovery through Synergizing Large Language Model and Data-Driven Reasoning. In *KDD*. 543–554.
- [6] Tian Gao and Qiang Ji. 2015. Local causal discovery of direct causes and effects. *NeurIPS* 28 (2015).
- [7] Xianjie Guo, Kui Yu, Lin Liu, Jiuyong Li, Jiye Liang, Fuyuan Cao, and Xindong Wu. 2024. Progressive skeleton learning for effective local-to-global causal structure learning. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [8] Kairong Han, Wenshuo Zhao, Ziyu Zhao, Junjian Ye, Lujia Pan, and Kun Kuang. 2025. CAT: Causal Attention Tuning For Injecting Fine-grained Causal Knowledge into Large Language Models. *arXiv preprint arXiv:2509.01535* (2025).
- [9] Zhijiang Jin, Yuen Chen, Felix Leeb, Luigi Gresele, Ojasv Kamal, Zhiheng Lyu, Kevin Blin, Fernando Gonzalez Adauto, Max Kleiman-Weiner, Mrinmaya Sachan, et al. 2023. Cladder: Assessing causal reasoning in language models. *NeurIPS* 36 (2023), 31038–31065.
- [10] Emre Kiciman, Robert Ness, Amit Sharma, and Chenhao Tan. 2023. Causal reasoning and large language models: Opening a new frontier for causality. *Transactions on Machine Learning Research* (2023).
- [11] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *NeurIPS* 35 (2022), 22199–22213.
- [12] Oscar Lithgow-Serrano, Vani Kanjirang, and Alessandro Antonucci. 2025. Causal Understanding by LLMs: The Role of Uncertainty. *arXiv preprint arXiv:2509.20088* (2025).
- [13] Chenxi Liu, Yongqiang Chen, Tongliang Liu, Mingming Gong, James Cheng, Bo Han, and Kun Zhang. 2024. Discovery of the hidden world with large language models. *NeurIPS* 37 (2024), 102307–102365.
- [14] Xiaoyu Liu, Paiheng Xu, Junda Wu, Jiaxin Yuan, Yifan Yang, Yuhang Zhou, Fuxiao Liu, Tianrui Guan, Haoliang Wang, Tong Yu, et al. 2025. Large language models and causal inference in collaboration: A comprehensive survey. *Findings of the Association for Computational Linguistics: NAACL 2025* (2025), 7668–7684.
- [15] Stephanie Long, Alexandre Piché, Valentina Zantedeschi, Tibor Schuster, and Alexandre Drouin. 2023. Causal Discovery with Language Models as Imperfect Experts. In *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*.
- [16] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. 2021. Toward causal representation learning. *Proc. IEEE* 109, 5 (2021), 612–634.
- [17] Peter Spirtes and Clark Glymour. 1991. An algorithm for fast recovery of sparse causal graphs. *Social science computer review* 9, 1 (1991), 62–72.
- [18] Yi-Kun Tang, Heyan Huang, Xuewen Shi, and Xian-Ling Mao. 2024. Beyond labels and topics: discovering causal relationships in neural topic modeling. In *WWW*. 4460–4469.
- [19] Ruibo Tu, Chao Ma, and Cheng Zhang. 2023. Causal-discovery performance of chatgpt in the context of neuropathic pain diagnosis. *arXiv preprint arXiv:2301.13819* (2023).
- [20] Aniket Vashishtha, Abbavaram Gowtham Reddy, Abhinav Kumar, Saketh Bachu, Vineeth N. Balasubramanian, and Amit Sharma. 2024. Causal Inference Using LLM-Guided Discovery.
- [21] Matthew J Vowles, Necati Cihan Camgoz, and Richard Bowden. 2022. D’y like dags? a survey on structure learning and causal discovery. *Comput. Surveys* 55, 4 (2022), 1–36.
- [22] Guangya Wan, Yuqi Wu, Mengxuan Hu, Zhixuan Chu, and Sheng Li. 2024. Bridging causal discovery and large language models: A comprehensive survey of integrative approaches and future directions. *CoRR* (2024).
- [23] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS* 35 (2022), 24824–24837.
- [24] Moritz Willig, Matej Zečević, Devendra Singh Dhami, and Kristian Kersting. 2022. Can foundation models talk causality? *arXiv preprint arXiv:2206.10591* (2022).
- [25] Xingyu Wu, Kui Yu, Jibin Wu, and Kay Chen Tan. 2025. LLM Cannot Discover Causality, and Should Be Restricted to Non-Decisional Support in Causal Discovery. *arXiv preprint arXiv:2506.00844* (2025).
- [26] Kui Yu, Xianjie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2020. Causality-based feature selection: Methods and evaluations. *ACM Computing Surveys (CSUR)* 53, 5 (2020), 1–36.
- [27] Matej Zečević, Moritz Willig, Devendra Singh Dhami, and Kristian Kersting. 2023. Causal parrots: Large language models may talk causality but are not causal. *arXiv preprint arXiv:2308.13067* (2023).
- [28] Cheng Zhang, Stefan Bauer, Paul Bennett, Jiangfeng Gao, Wenbo Gong, Agrin Hilmkil, Joel Jennings, Chao Ma, Tom Minka, Nick Pawlowski, et al. 2023. Understanding causality with large language models: Feasibility and opportunities. *arXiv preprint arXiv:2304.05524* (2023).
- [29] LYU Zhiheng, Zhijiang Jin, Rada Mihalcea, Mrinmaya Sachan, and Bernhard Schölkopf. 2022. Can large language models distinguish cause from effect?. In *UAI 2022 Workshop on Causal Representation Learning*.

A Benchmark Data Construction and Evaluation Framework

To address the lack of high-quality textual datasets for causal discovery methods, we design a systematic benchmark data construction framework. We employ ancestral sampling to generate structured datasets, starting from root node values, ensuring that the generated data conforms to predefined causal structures and probability distributions.

Probability Table Acquisition. Given a causal network with N variables, we represent its structure using a directed acyclic graph $G = (V, E)$, where $V = \{V_1, V_2, \dots, V_N\}$ is the variable set. Each variable V_i can have different value spaces, such as binary variables $V_i \in \{0, 1\}$ or multivariate variables $V_i \in \{1, 2, \dots, k\}$.

For conditional probability model construction, we adopt two approaches:

1) **Using Existing Probability Tables.** If the causal graph under study already has validated Conditional Probability Tables (CPTs) through expert knowledge or historical data, we directly use these established parameters:

$$P(V_i | \text{Pa}(V_i)) = \text{CPT}. \quad (13)$$

2) **Generating Random Probability Tables.** When existing CPTs are unavailable (e.g., in newly constructed or theoretical causal graphs), we generate reasonable conditional probability distributions based on Dirichlet priors. The specific procedure is as follows:

First, determine the number of parent state combinations:

$$n_{\text{rows}} = \prod_{V_j \in \text{Pa}(V_i)} |V_j|, \quad (14)$$

where $|V_j|$ represents the number of states of variable V_j .

Then, for each parent state combination, sample conditional probability vectors from the Dirichlet distribution:

$$\mathbf{p} = (p_1, p_2, \dots, p_{r_i}) \sim \text{Dirichlet}(\alpha_1, \alpha_2, \dots, \alpha_{r_i}), \quad (15)$$

where r_i is the number of states of variable V_i , typically using symmetric hyperparameters $\alpha_1 = \alpha_2 = \dots = \alpha_{r_i} = 1$ by default. The hyperparameter α primarily controls the concentration of generated conditional probability distributions. By adjusting α values, we can generate datasets with varying certainty and noise levels.

Table 3: Comparison of three pipelines (DATA+CoT, COAT, CAST) across datasets and LLMs (additional models) for factor extraction, Factor Assignment, and Causal Feature Learning. Best results are in bold.

Dataset	LLM	Method	Factor Extraction			Factor Assignment	Causal Feature Learning		
			Recall	Precision	F1		Recall	Precision	F1
AG	Deepseek-r1:8b	DATA+CoT	84.00	66.03	73.94	/	68.00	61.75	64.72
		COAT	90.00	65.99	76.15	67.34	70.00	78.71	74.10
		CAST(ours)	92.00	65.87	76.77	75.50	72.00	81.83	76.60
	Deepseek-r1:32b	DATA+CoT	96.00	56.30	70.98	/	88.00	72.60	79.56
		COAT	96.00	57.90	72.23	82.05	90.00	80.19	84.81
		CAST(ours)	100.0	58.08	73.48	88.12	90.00	86.00	87.95
	Qwen3:8b	DATA+CoT	72.00	42.00	53.05	/	72.00	42.67	53.58
		COAT	58.00	56.45	57.21	87.49	54.00	66.83	59.73
		CAST(ours)	72.00	67.38	69.61	87.78	62.00	87.17	72.46
	Qwen3:14b	DATA+CoT	96.00	49.59	65.40	/	84.00	54.72	66.27
		COAT	76.00	57.89	65.72	88.95	68.00	77.50	72.44
		CAST(ours)	78.00	69.79	73.67	89.76	72.00	85.95	78.36
LD	Deepseek-r1:8b	DATA+CoT	84.00	38.99	53.26	/	52.00	47.68	49.74
		COAT	52.00	46.34	49.01	76.77	52.00	55.33	53.61
		CAST(ours)	86.00	44.96	59.05	94.72	82.00	64.10	71.95
	Deepseek-r1:32b	DATA+CoT	88.00	56.19	68.59	/	36.00	48.00	41.14
		COAT	70.00	45.15	54.89	97.88	66.00	70.00	67.94
		CAST(ours)	98.00	49.65	65.91	99.66	98.00	79.29	87.66
	Qwen3:8b	DATA+CoT	100.00	45.76	62.79	/	96.00	54.44	69.48
		COAT	92.00	45.32	60.73	97.18	92.00	72.90	81.34
		CAST(ours)	98.00	50.10	66.30	99.19	98.00	70.48	81.99
	Qwen3:14b	DATA+CoT	100.00	50.00	66.67	/	76.00	73.67	74.82
		COAT	100.00	38.71	55.81	99.15	86.00	67.39	75.57
		CAST(ours)	98.00	47.14	63.66	99.98	90.00	72.02	80.01
WM	Deepseek-r1:8b	DATA+CoT	56.67	21.91	31.60	/	30.00	27.83	28.88
		COAT	53.33	35.69	42.76	62.22	41.88	43.33	42.59
		CAST(ours)	70.00	67.71	68.84	67.02	60.00	64.10	61.98
	Deepseek-r1:32b	DATA+CoT	70.00	52.54	60.03	/	30.00	53.33	38.40
		COAT	60.00	32.08	41.81	74.80	43.33	29.02	34.76
		CAST(ours)	70.00	56.35	62.44	79.13	60.00	70.33	64.76
	Qwen3:8b	DATA+CoT	60.00	20.78	30.87	/	40.00	18.94	25.71
		COAT	86.67	44.35	58.68	73.10	66.83	59.88	63.16
		CAST(ours)	90.00	74.62	81.59	75.79	83.33	76.24	79.63
	Qwen3:14b	DATA+CoT	63.33	21.04	31.59	/	43.33	19.81	27.19
		COAT	80.00	31.96	45.67	80.10	56.67	65.71	60.86
		CAST(ours)	86.67	68.81	76.71	81.62	73.33	74.29	73.81

Finally, construct the complete CPT and ensure probability normalization:

$$\sum_{k=1}^{r_i} p_{k,j} = 1, \quad \forall j = 1, 2, \dots, n_{\text{rows}}. \quad (16)$$

Structured Data Generation. After obtaining the complete probability model, we employ the Ancestral Sampling method to generate structured datasets. This method strictly follows the topological order of the causal graph, ensuring that the generated data conforms to the predefined causal mechanisms.

First, generate values for all root nodes:

$$x_i^{(m)} \sim P(V_i), \quad m = 1, 2, \dots, M, \quad (17)$$

where M is the sample size.

Then, following the topological ordering of the network, sequentially generate values for each child node:

$$x_j^{(m)} \sim P(V_j | \text{Pa}(V_j) = \mathbf{x}_{\text{Pa}(V_j)}^{(m)}), \quad (18)$$

where $\mathbf{x}_{\text{Pa}(V_j)}^{(m)}$ represents the values of the parent nodes of V_j in the m -th sample.

Two-level Verification Mechanism. To ensure the reliability of the generated data, we employ a two-level verification mechanism:

Level 1: Conditional Independence Verification. : At this stage, we use the Kernel Conditional Independence Test (KCI) to verify whether the generated data contains the expected direct causal relationships. For each direct causal edge present in the causal graph, we test whether the corresponding two variables are significantly dependent given an empty conditioning set:

$$\text{KCI}(X, Y | \emptyset) < \alpha, \quad \alpha = 0.01. \quad (19)$$

This step ensures that the generated dataset genuinely reflects the predefined causal structure.

Level 2: Causal Structure Learning Verification. : After passing the conditional independence verification, we further employ causal structure learning algorithms such as FCI to learn causal structures from the generated data and verify their consistency with the true structure:

$$G_{\text{learned}} = \text{FCI}(X, \alpha_{\text{FCI}}), \quad (20)$$

where α_{FCI} is the significance level used by the FCI algorithm. We systematically check the existence and direction of key causal relationships in the learned graph and strictly exclude any spurious reverse edges. If the structure learned fails verification, we regenerate the data and repeat the above process until a dataset that meets the requirements is obtained.

Textual Data Generation. The verified structured data is used to generate corresponding textual descriptions. We employ an LLM-based text generation framework, establishing a mapping relationship between feature values and descriptive text for each sample in the structured dataset:

$$f : (j, i_j) \rightarrow \text{Description}_j. \quad (21)$$

Based on the mapped feature descriptions, we construct structured prompt templates to guide the LLM in generating appropriate text, and conduct manual verification to ensure the correctness of the final generated textual data and high-quality linguistic expression.

This framework, through rigorous probability modeling, data generation, and verification mechanisms, ensures that the constructed datasets satisfy both the theoretical constraints of the causal graph and support subsequent textual causal discovery research, providing a reliable foundation for evaluating causal discovery methods.

B Experimental Results

Compute-Efficient Effectiveness of CAST Across LLM Scales. Table 3 demonstrates that CAST consistently surpasses DATA+CoT and COAT across three datasets and a range of LLM sizes, with particularly strong gains on smaller LLMs, indicating low compute requirements. Concretely, with 8b models CAST narrows or closes the gap to larger backbones—for example, on AG/Qwen3-8b it achieves extraction F1 69.61% vs 57.21% (COAT) and causal F1 72.46% vs 59.73%; on WM/Qwen3-8b it attains causal F1 79.63%; and on LD/Deepseek-8b it raises assignment accuracy to 94.72% vs 76.77% (COAT), with AG/Deepseek-8b also improving to 75.50% vs 67.34%. Similar trends persist at 32b, but the magnitude of improvements on 8b highlights that CAST’s advantages derive from its stage-wise algorithmic design rather than model scale, making it attractive for resource-constrained deployments.

Causal Structure Learning Performance. As illustrated in Figure 4, the causal structures learned by CAST are consistently closer to the ground truth and more parsimonious than those produced by COAT. On the AG dataset, CAST stably recovers the correct parent set of score—including size, aroma, and taste—and accurately preserves the directed edge score $\rightarrow$ market_potential. In contrast, COAT frequently introduces extraneous variables (e.g., freshness, texture, consumer_satisfaction, risk), leading to fragmented graph structures and often misoriented edges around score. On the LD dataset, CAST fully reconstructs the core causal pathway anxiety/stress $\rightarrow$ smoking $\rightarrow$ lung_disease $\rightarrow$ {cough, fatigue}, while correctly retaining key factors such as hereditary influences on attention_disorder. COAT, however, produces disconnected components (e.g., occupation, environment) and long-range links unsupported by the ground truth. On the real-world WM dataset, CAST accurately identifies all key factors influencing the target variable (e.g., ripeness, freshness, and taste), whereas COAT not only extracts numerous irrelevant variables but also incorrectly infers the directions of all causal links affecting the target variable. Experimental results show that across base models of various parameter scales (14b, 20b and 32b), the causal graphs learned by CAST consistently exhibit more compact Markov blankets and clearer v-structures, demonstrating significant advantages even on challenging real-world datasets. This fully

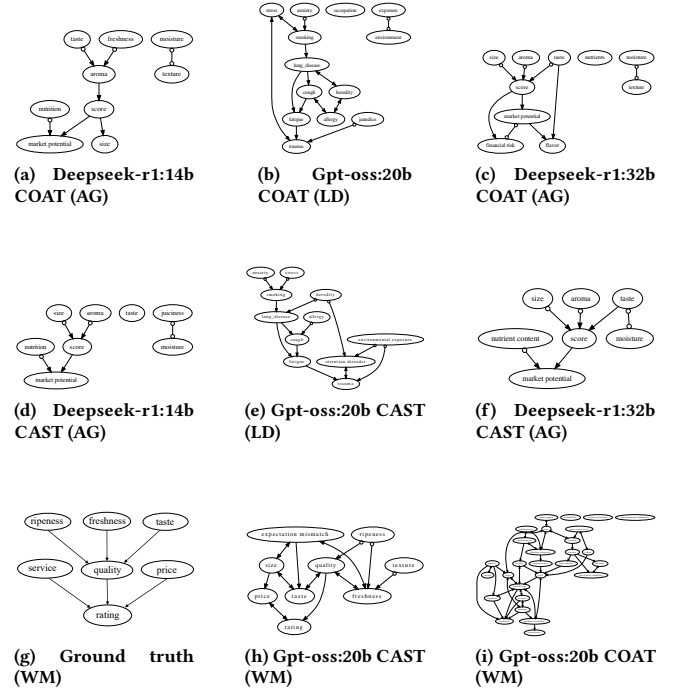


Figure 4: Causal graphs discovered by COAT and CAST (ours) across different LLMs.

validates CAST’s strong robustness to the choice of base models and its clear superiority over the COAT method.

Runtime Analysis. Table 4 compares the running times of the COAT and CAST methods using different models on the Apple-Gastronome (AG) dataset. The experiments were conducted on a hardware platform equipped with an Intel Core i9-10900F CPU, 32GB RAM, and an NVIDIA GeForce RTX 3060 GPU (12GB VRAM), running Windows 11 and Python 3.12. The primary computational overhead for both methods comes from large language model (LLM) calls. Although CAST’s multi-round voting strategy increases time overhead, its early-termination mechanism effectively controls the overall runtime. For instance, with the Deepseek-r1:8b model, the runtime increases from 859.31 seconds for COAT to 1141.44 seconds for CAST. Interestingly, with the Deepseek-v3-0324 model, CAST (1220.44 seconds) runs even faster than COAT (1343.34 seconds), primarily due to CAST’s faster convergence. Overall, the moderate increase in runtime for CAST is acceptable, and it achieves an average improvement of 14.9% in F1 score across all models, demonstrating a good balance between efficiency and effectiveness.

Downstream Causal Effect Estimation. To further assess end-to-end utility, Table 5 and Table 6 compares causal effect estimation based on the structured outputs of the two methods. On Deepseek-r1:32b, the absolute error of the Average Treatment Effect (ATE) decreases from 0.067 to 0.036 ($\approx 46\%$ reduction), PEHE drops from 0.204 to 0.150, and Pearson r increases from 0.448 to 0.585. Similar trends appear on Deepseek-r1:14b (ATE 0.086 $\rightarrow$ 0.046; $\approx 46.5\%$ reduction). A mild trade-off appears on Qwen3:8b: ATE increases

Table 4: Running time (in seconds) of COAT and CAST (ours) methods with different models on the AG dataset.

	Deepseek-r1:8b	Deepseek-r1:14b	Deepseek-v3-0324	Qwen-plus	Qwen3-max
COAT	859.31	2709.20	1343.34	558.41	917.05
CAST	1141.44	3384.34	1220.44	788.21	1394.05

Table 5: AppleGastronome dataset causal effect estimation results (ATE absolute error, Pearson correlation, PEHE). Best results are in bold.

LLM	Method	ATE ↓	Pearson r ↑	PEHE ↓
Deepseek-r1:8b	COAT	0.111 ± 0.071	0.291 ± 0.199	0.234 ± 0.057
	CAST	0.060 ± 0.019	0.417 ± 0.135	0.179 ± 0.022
Deepseek-r1:14b	COAT	0.086 ± 0.026	0.414 ± 0.124	0.216 ± 0.036
	CAST	0.046 ± 0.018	0.512 ± 0.132	0.159 ± 0.015
Deepseek-r1:32b	COAT	0.067 ± 0.013	0.448 ± 0.117	0.204 ± 0.038
	CAST	0.036 ± 0.013	0.585 ± 0.093	0.150 ± 0.018
Deepseek-v3-0324	COAT	0.086 ± 0.015	0.533 ± 0.067	0.196 ± 0.024
	CAST	0.076 ± 0.024	0.617 ± 0.076	0.162 ± 0.029
Qwen3:8b	COAT	0.054 ± 0.021	0.508 ± 0.114	0.180 ± 0.030
	CAST	0.065 ± 0.019	0.458 ± 0.103	0.172 ± 0.021
Qwen3:14b	COAT	0.072 ± 0.071	0.504 ± 0.129	0.174 ± 0.061
	CAST	0.051 ± 0.023	0.525 ± 0.111	0.153 ± 0.024
Qwen3:32b	COAT	0.078 ± 0.042	0.413 ± 0.169	0.207 ± 0.035
	CAST	0.054 ± 0.019	0.523 ± 0.163	0.156 ± 0.023
Qwen-plus	COAT	0.047 ± 0.005	0.689 ± 0.035	0.134 ± 0.008
	CAST	0.047 ± 0.008	0.716 ± 0.038	0.126 ± 0.006
Gpt-oss:20b	COAT	0.067 ± 0.015	0.534 ± 0.104	0.195 ± 0.031
	CAST	0.045 ± 0.018	0.545 ± 0.154	0.150 ± 0.025

Table 6: LungDisease dataset causal effect estimation results (ATE absolute error, Pearson correlation, PEHE). Better results (between COAT and CAST) are shown in bold.

LLM	Method	ATE ↓	Pearson r ↑	PEHE ↓
Deepseek-r1:8b	COAT	0.053 ± 0.027	0.576 ± 0.236	0.174 ± 0.049
	CAST	0.042 ± 0.030	0.795 ± 0.132	0.130 ± 0.042
Deepseek-r1:14b	COAT	0.039 ± 0.044	0.747 ± 0.219	0.145 ± 0.065
	CAST	0.019 ± 0.019	0.888 ± 0.037	0.093 ± 0.020
Deepseek-r1:32b	COAT	0.045 ± 0.019	0.802 ± 0.150	0.134 ± 0.041
	CAST	0.009 ± 0.012	0.959 ± 0.025	0.057 ± 0.020
Deepseek-v3-0324	COAT	0.003 ± 0.003	0.985 ± 0.009	0.031 ± 0.010
	CAST	0.004 ± 0.005	0.973 ± 0.036	0.037 ± 0.024
Qwen3:8b	COAT	0.017 ± 0.014	0.910 ± 0.057	0.086 ± 0.027
	CAST	0.006 ± 0.006	0.952 ± 0.023	0.062 ± 0.014
Qwen3:14b	COAT	0.032 ± 0.041	0.903 ± 0.094	0.092 ± 0.064
	CAST	0.019 ± 0.029	0.891 ± 0.131	0.084 ± 0.058
Qwen3:32b	COAT	0.004 ± 0.002	0.961 ± 0.031	0.052 ± 0.025
	CAST	0.003 ± 0.001	0.984 ± 0.011	0.034 ± 0.010
Qwen-plus	COAT	0.006 ± 0.013	0.970 ± 0.046	0.041 ± 0.028
	CAST	0.005 ± 0.005	0.971 ± 0.026	0.046 ± 0.019
Gpt-oss:20b	COAT	0.014 ± 0.013	0.908 ± 0.040	0.092 ± 0.028
	CAST	0.007 ± 0.003	0.949 ± 0.024	0.063 ± 0.015

slightly (0.054 → 0.065), while PEHE still improves (0.180 → 0.172) and correlation remains competitive—consistent with heightened sensitivity to assignment uncertainty at minimal parameter scales; our feedback loop mitigates this but cannot fully eliminate it in the smallest models.

Prompt Design for Factor Extraction and Assignment. To ensure the reliability and interpretability of factor extraction, two structured prompts were designed for medical text analysis. The Factor Extraction Prompt (Figure 5) instructs the model to extract novel, medically meaningful, and semantically independent base concepts

```
system_content = f"""
You are a medical research expert specializing in extracting new core factors that may be relevant to lung disease, based on the given clinical reports or patient descriptions. Each factor should be expressed as a single term or concise phrase.
Note:
0. Extract ONLY BASE CONCEPTS (remove all adjectives, modifiers and qualifiers).
1. Do not repeat existing factors or include meaningless ones such as "factor", "new factor1", etc.
2. Extracted factors should be general, medically meaningful, and semantically independent.
Please strictly use the following JSON format for the output:
{{
  "factors": ["new factor1", "new factor2", ...]
}}
```

Figure 5: Factor Extraction Prompt

```
prompt_template = {
  "template": """
  [Role Definition]
  You are a clinical text analysis expert. Your task is to annotate whether certain medical-related factors are explicitly mentioned in each review.

  # Annotation Guidelines
  ## Labeling Criteria
  - YES (1): The text explicitly confirms the presence of the factor.
    - e.g., "The patient reports persistent anxiety", "Has a long history of smoking"
  - NO (-1): The text explicitly denies the factor OR states its absence.
  - Explicit denial: "No signs of anxiety", "Non-smoker"
  - Absence statement: "low stress levels", "stress is absent", "no evidence of stress"
  - Not Mentioned (0):

  # Current Task
  ## Factor List
  {factors}
  ## Reviews to Annotate
  {reviews}

  Please provide the output in the following JSON format:
  """,
  "factors": factor_columns,
  "criteria": criteria
}
```

Figure 6: Factor Assignment Prompt

related to lung disease while avoiding redundant or overly specific terms. The Factor Assignment Prompt (Figure 6) provides explicit labeling criteria—YES (1), NO (−1), and Not Mentioned (0)—for determining the presence or absence of each factor in clinical texts. Together, these prompts establish a standardized and interpretable framework for automated factor extraction and annotation.